

Research Statement

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I. VISION: FROM ROBUST SLAM TO LIFELONG SPATIAL MAPPING

A. Background

As robots transition from structured laboratories to field environments such as disaster sites, ports, and houses, they must navigate safely under severe uncertainty [1]. Simultaneous Localization and Mapping (SLAM) [2] has emerged as a foundation for spatial perception, and mature metric SLAM already enables commercial drones and AR glasses [3]. However, general-purpose robots demand more: perception must move beyond geometric mapping towards **Spatial AI** [4], encompassing semantic understanding, object relation inference, and long-term map maintenance. At the frontier of this evolution lies the concept of **World Models**, internal representations that enable robots to predict, plan, and imagine future states before acting, bridging the gap between passive perception and active intelligence.

Driven by this paradigm shift, my research centers on developing a *lifelong spatial mapping* system, serving as a dedicated hippocampus for robotics [5]. The concept of lifelong emphasizes a robot's ability to execute a sequence of tasks over an extended temporal and spatial lifespan [6]–[8]. Unlike standard embodied tasks that execute isolated commands (e.g., finding a kitchen), lifelong autonomy requires handling multiple sequential objectives. For instance, transitioning from finding a coffee cup to locating a tea cup requires recognizing that functionally similar items likely share the same location. Utilizing past experiences to infer future states fundamentally demands robots to memorize, reflect, and consolidate knowledge to improve problem-solving in analogous situations.

My research trajectory parallels this evolution, progressing from robust low-level geometric perception to high-level lifelong spatial mapping. Initially, I enhanced mobile robot perception by leveraging multi-LiDAR configurations [9]–[11], with novel extrinsic calibration, data fusion, and uncertainty modeling to achieve stable state estimation in complex environments. I further introduced the FusionPortable V1-V2 benchmarks [12], [13] to evaluate the generalizability of multi-sensor SLAM algorithms, alongside high-precision heterogeneous sensor calibration and fusion methods [14], [15]. However, these systems primarily prioritized geometric precision over long-term maintenance. My current research advances towards a scalable lifelong mapping paradigm [16]–[18]: a framework offering continuous adaptation, semantic understanding, and crowdsourced map updates.

B. Research Questions

Overall, my past and current research aim to address the following two key scientific questions:

Research Question 1 (RQ1): *How can we model and mitigate multi-sensor extrinsic uncertainties and geometric degeneracies to guarantee state estimation observability in unstructured environments?*

Research Question 2 (RQ2): *How can we achieve spatial mapping by leveraging crowdsourced updates and globally-consistent map fusion for lifelong robot navigation?*

As illustrated in Fig. 1, addressing these questions involves five key objectives:

- *Large-scale mapping:* Scalable, resource-efficient maps that remain consistent as environments grow.
- *Appearance change:* Robust perception under lighting, weather, and seasonal variations.
- *Mobility:* Adaptive traversability estimation and planning on diverse terrains.
- *Scene understanding:* Semantic interpretation of objects, human intent, and social context.
- *Robustness:* Fault tolerance against sensor degradation, interference, and hardware failures.

II. PAST AND CURRENT RESEARCH

A. Problem Statement

Simultaneous Localization and Mapping (SLAM) estimates robot states while building a 3D map from sensor data. A typical pipeline has a *front end* (feature extraction and data association) and a *back end* (optimization over a geometric model). The estimation problem can be written as

$$\hat{\mathcal{X}} = \arg \min_{\mathcal{X}} f(\mathcal{X}; \mathcal{Z}), \quad (1)$$

where $f(\cdot)$ is the objective and $\mathcal{Z}$ is the set of measurements. With multiple sensors, SLAM couples with extrinsic calibration:

$$\hat{\mathcal{X}}, \hat{\mathcal{Y}} = \arg \min_{\mathcal{X}, \mathcal{Y}} f(\mathcal{X}, \mathcal{Y}; \mathcal{Z}), \quad (2)$$

where $\mathcal{Y} = \{\mathbf{y}_i\}_{i=1}^N$ are sensor extrinsics. This yields a classic “chicken-and-egg” coupling: accurate fusion needs good calibration, while optimal calibration requires reliable SLAM. The challenge is central to modern autonomous systems, including mobile robots, mapping rigs, and AR/VR devices, whose robustness hinges on multi-sensor fusion.



Fig. 1. Real-world challenges in navigation: large-scale localization and mapping, appearance change, mobility, scene understanding, and robustness.

TABLE I
LIST OF REPRESENTATIVE PUBLICATIONS.

Venue	Count
The International Journal of Robotics Research (IJRR) [13]	1
IEEE Transactions on Robotics (TRO) [11], [17], [19]	3
IEEE/ASME Transactions on Mechatronics (TMech) [14], [20]–[22]	4
IEEE Transactions on Automation Science and Engineering (TASE) [15]	1
IEEE Robotics and Automation Magazine (RAM)	1
IEEE International Conference on Robotics and Automation (ICRA) [16], [23]–[30]	9
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) [9], [10], [12], [31]	4
Neural Information Processing Systems (NeurIPS) [32]	1
ACM Conference on Embedded Networked Sensor Systems (SenSys) [33]	1

B. Robust Geometric Perception (RQ1)

To address **RQ1**, my research establishes an advanced multi-LiDAR perception system for autonomous robots. While single LiDARs commonly suffer from limited observability in structureless environments, multi-LiDAR setups significantly expand the sensing range and point density to eliminate perceptual blind spots. I systematically contributed approaches in extrinsic calibration, multi-LiDAR data fusion, uncertainty-aware mapping, and multi-sensor benchmarking. All datasets and code are publicly available, and the resulting contributions have been widely adopted as standard baselines across both academic research and industrial robotic systems.

- **Online Extrinsic Calibration.** Conventional multi-sensor systems typically assume static, offline-calibrated extrinsic parameters, making them vulnerable to mechanical vibrations and thermal deformations [34]. However, online calibration methods often yield suboptimal extrinsics when environmental or motion constraints are not fully observable [35]. To overcome this, I developed automatic self-calibration methods based on appearance and motion patterns [9], [36], with online convergence identification to lock accurate extrinsics. To further handle cm-level calibration errors and unexpected extrinsic drift over time, I explicitly model extrinsic perturbations within the multi-LiDAR system [11].
- **M-LOAM: Multi-LiDAR Odometry and Mapping.** While calibration addresses inter-sensor alignment, residual uncertainties must be propagated into the mapping pipeline. Conventional LiDAR SLAM systems [37] rely on single-sensor setups and do not incorporate such uncertainty modeling into map construction. I developed M-LOAM [11], a tightly-coupled framework that explicitly models sensor noise, extrinsic errors, and pose estimation degeneracies within a sliding-window optimization graph. I further proposed an uncertainty-aware feature selection strategy [24] guarantees geometric observability even in highly degenerate scenes, achieving localization drift under $30cm$ over $700m$ indoors and less than 2.5% over $3.4km$ in urban settings.
- **MLOD: Multi-LiDAR 3D Object Detection.** This uncertainty-aware paradigm was further extended to perception via the proposed MLOD [10] method. Existing detectors [38] require perfectly calibrated setups and degrade severely under extrinsic perturbations. MLOD propagates an uncertainty prior onto the input point clouds and employs a two-stage fusion network to incorporate spatial errors, improving detection accuracy by over 10% in mAP under high perturbations.
- **FusionPortable V1-V2: Multi-Sensor SLAM Benchmark.** To systematically evaluate the generalizability of multi-sensor SLAM algorithms, I proposed the FusionPortable benchmark series [12], [13], alongside high-precision heterogeneous sensor calibration and fusion methods [14], [15]. FusionPortable provides a comprehensive multi-sensor suite (LiDARs, stereo/event cameras, IMUs, GPS) deployed across diverse platforms (wheeled, legged, handheld, vehicular) and environments (campuses, underground areas, highways). FusionPortableV2 scales to 27 sequences, $38.7km$ total path length, and 12+ environment types with ground-truth trajectories and RGB point cloud maps, serving as a challenging testbed for evaluating SLAM robustness and generalization.

C. Lifelong Spatial Mapping (RQ2)

While the above research ensures reliable single-session state estimation, the utility of such maps for lifelong autonomy remains an open challenge. My current research addresses **RQ2**: achieving large-scale, multi-session visual mapping via crowdsourced updates, tackling two key challenges: long-term data association via generalized place recognition (PR), and the design of lightweight, reusable map representations.

- **Generalized Place Recognition Survey.** I co-authored a comprehensive survey on generalized PR [17]. Prior surveys [39]–[41] predominantly evaluate PR as an isolated vision task, overlooking its integration into lifelong mapping pipelines. We systematically categorize the latest approaches, analyze existing bottlenecks, and identify core challenges from the perspective of scalable spatial memory. Building on these insights, I proposed a flexible sequence matching method based on dynamic programming [18] that robustly resolves data association across multiple, randomly crowdsourced trajectories.
- **LiteVLoc and OpenNavMap: Lifelong Topometric Mapping.** I developed LiteVLoc [16] and OpenNavMap [18], a lightweight, structure-free topometric mapping system for large-scale, dynamic environments. Traditional SfM approaches [42] incur massive storage costs and fail under severe viewpoint and illumination changes, while absolute pose regression models [43], [44] generalize poorly to unseen environments. My approach leverages 3D geometric foundation models [45] to construct a sparse topo-metric representation, reducing equipment costs and enabling crowdsourced map updates. It achieves an average translation error of $0.62m$ while maintaining global consistency across $15km$ of multi-session data with map-merging absolute trajectory error below $3m$. The system was validated using crowdsourced data from AR glasses, vehicles, smartphones, and Google Street View cameras, and the resulting map was deployed on a quadruped robot for fully autonomous image-goal navigation.

D. Technology Transfer and Impact

- **MATLAB Integration.** My proposed calibration method was incorporated into MATLAB’s LiDAR Toolbox [46], making it accessible to the broader engineering community.
- **Open-Source Adoption.** The M-LOAM framework has garnered 500+ GitHub stars and is widely adopted as a baseline in multi-LiDAR SLAM research [47]. The FusionPortable dataset is actively maintained and used in many research projects.
- **FusionPortable Benchmark.** The FusionPortable V1-V2 datasets [12], [13] provide the SoTA multi-sensor SLAM benchmark. The benchmark has been cited and adopted by research groups at leading institutions including ETH Zurich, Stanford, UMich, and ICL. The sensor suite hardware has been further developed into a product-grade multi-sensor device for broader deployment.
- **Industrial Deployment.** Our technology has been transferred to industrial autonomous driving products and contributed to obtaining the first moving permit in Hong Kong for autonomous vehicles.
- **Publication.** These works result in 1 IJRR, 3 TRO, 4 T-Mech, and 13 ICRA/IROS publications, as summarized in Table I.

III. FUTURE RESEARCH AGENDA (FIVE-YEAR PLAN)

A. Motivation

My past research has established two complementary threads: robust multi-sensor state estimation, exemplified by M-LOAM and FusionPortable, and lightweight crowdsourced mapping, exemplified by LiteVLoc and OpenNavMap. Together, they point to a broader unresolved problem: for robots to operate over months or years in the physical world, a map cannot remain a static geometric representation; it must become a persistent physical world model that supports perception, memory, retrieval, and reasoning. This need is driven by three practical limitations of current systems. First, SLAM remains fragile under long-term environmental changes such as moved objects, renovated layouts, and seasonal appearance shifts, with little ability for self-updating or selective forgetting. Second, dominant vision-language-action (VLA) models encode physical knowledge only implicitly, whereas manipulation depends on explicit priors such as material properties, articulation types, and motion ranges. Third, progress in multimodal embodied perception is limited by the scarcity of large-scale tactile data; my planned collaboration with Daimeng Robotics will provide an industry-scale visuo-tactile grasping dataset to address this bottleneck.

The central **research question** is therefore: *How can we build accurate, lightweight, and dynamically consistent lifelong multi-modal 3D memory for mobile manipulators, so that it serves as an effective physical prior across tasks and environments?*

B. Proposed Works

I will pursue two tightly connected directions: lifelong multimodal 3D memory as the main research thrust, and memory-driven embodied action as the validation pathway. The first direction will contribute the core representation, learning, and system foundations; the second will quantify how much structured memory improves downstream robot behavior and will feed interaction experience back into memory updates.

1) **Direction 1: Scalable Lifelong Spatial Memory:** I will extend my current mapping work into a lifelong multimodal 3D memory system that explicitly represents geometry, semantics, physical attributes, and temporal change, while remaining deployable in large-scale real environments.

- *Grounded multimodal 3D scene graph construction.* Build a hierarchical 3D scene-graph representation in which each node corresponds to an object or region instance and stores visual appearance, semantic category, tactile material cues, and, when relevant, kinematic parameters such as joint type and motion range. This extends my prior geometric mapping foundations with explicit physical attributes needed for interaction, while remaining general enough to support higher-level reasoning and confidence-aware fusion.
- *Dynamic consistency and hierarchical memory compression.* Develop a memory architecture that distinguishes between frequently interacted objects and slowly varying background structure. Active object nodes will retain precise explicit representations, while stable regions will be compressed into lower-dimensional latent summaries to reduce storage cost without losing essential environmental statistics. I will incorporate 3D spatial constraints as structured regularization during continual learning so that the system can update over time without catastrophic forgetting.
- *Task-driven retrieval and physical reasoning.* Design retrieval interfaces that expose structured memory to downstream navigation and manipulation modules. For example, when the robot encounters an unseen cabinet door, it should be able to retrieve related interaction histories from similar articulated objects and infer likely opening direction, material, and force requirements. Likewise, stored priors on traversability, slope, and surface properties should improve long-horizon planning reliability. This makes the memory system a usable physical prior rather than a passive archive.
- *Long-horizon field validation.* Validate the memory system in industrial inspection, campus navigation, and mock disaster-response sites, measuring memory compression, multi-session consistency, and robustness under environmental changes. These trials will also test decentralized updates across heterogeneous devices, continuing the collaborative mapping line initiated in OpenNavMap.

2) **Direction 2: From Spatial Memory to Embodied Action:** Rather than developing a new end-to-end action foundation model, I will use representative open VLA backbones to quantify how structured memory improves embodied behavior and how interaction outcomes should update the memory itself.

- *Memory-guided semantic navigation.* Use the multimodal scene graph as a long-horizon prior for open-vocabulary navigation tasks such as locating functionally or materially specified targets. Stored traversability and kinematic priors will inform planning, allowing direct measurement of gains in success rate and path efficiency over memory-free baselines.
- *Contact-rich manipulation with physical priors.* Inject object-level material and articulation attributes from memory into VLA policies for grasping, pulling, turning, and other contact-rich skills. The visuo-tactile data collected with Daimeng Robotics will support this line by enabling robust estimation of contact geometry and physical properties, while online interaction outcomes will update the corresponding memory nodes.
- *Cross-platform generalization.* Evaluate whether the same memory representation transfers across a wheeled mobile manipulator and a legged manipulator, thereby testing embodiment-agnostic physical memory. Performance will be measured through task success, adaptation speed, and robustness across industrial, campus, and disaster-response scenarios.

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